Revising the n_TOF PSA configuration for the X17 EAR2 campaign

What the original production reconstruction gets wrong, and what the revision measures instead

X17 / DREAM group

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Abstract

The official Pulse Shape Analysis configuration for the 2026 X17 EAR2 campaign mis-identifies the γ-flash on the plastic scintillators in **37–85 % of bunches**, and times the flash on the SiPM walls from the transient of their own protection gate rather than from the flash itself. Since every hit's physics time is tof $- t_{\text{flash}}$, the first defect shifts whole sets of hits by up to 11.6 μs, and the second puts a fixed $\sim$350 ns offset between detectors that should agree. Together they made a hardware wall $\wedge$ plastic coincidence look as though the plastic fired only half the time.

Both are configuration faults, not detector faults: the raw waveforms contain the flash, unmissably, in every bunch. This note documents the revision, the measurements behind each change, and what it costs. On the reference run the flash mis-identification goes to **0.0 % on 12 of 13 trees**, the per-arm coincidence offsets go from $-362/+20/-333/-336$ ns to $+1.5/+2.0/+1.0/-2.0$ ns, and our DREAM coincidence matcher improves from 93.7 % to **96.4 %** efficient — the latter measured with our offline repair switched off, so it tests the processing alone.

1 Why this was worth chasing

The X17 measurement pairs the n_TOF scintillator array with the DREAM detector. Its trigger is a hardware AND of a SiPM wall and a plastic scintillator in the same arm, so every triggered event must have a plastic partner. In the official reconstruction only about half of them did.

The explanation is not in the detectors. It is that $t_{\text{since flash}} = \text{tof} - t_{\text{flash}}$ is only as good as the PSA's flash identification, and on the plastics that identification fails in most bunches.

2 Defect 1: the plastic flash is mis-identified

The plastic row of the UserInput sets `G-FLASH THRESHOLD = 50.` with no lower time limit. The plastic flash saturates the digitiser — it drives the signal from a baseline near 30 800 past zero — so 50 channels is about 0.2 % of the feature it is meant to find. Whatever noise excursion happens first in the 11.6 μs before the flash wins instead.

The fix is to give the finder a threshold matched to the feature and a lower time limit: `G-FLASH THRESHOLD = 2000/1e4`. The flash sits at 11.63 μs in every bunch by hardware, so refusing to look before 10 μs removes the entire failure mode. The plateau is wide — 500, 2000 and 5000 give identical results on 851 channel–bunch combinations — so the choice is not delicate.

3 Defect 2: the walls time their own protection gate

The SiPM wall signal is blanked by a $\sim$1 μs protection gate around the flash. The walls therefore never see the flash directly. What the digitiser records, in order, is: the gate *closing* at 11.24 μs; a

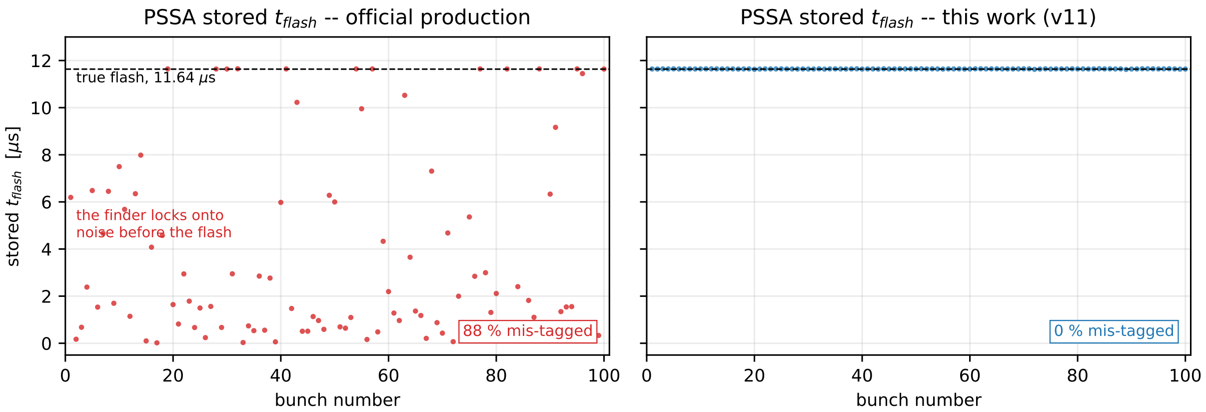

Figure 1: Stored t_{flash} for each bunch of tree PSSA. *Left*: the official production. The true flash is the dashed line at 11.64 μs; most bunches are tagged somewhere in the noise before it, some by more than 11 μs. *Right*: the same bunches after the revision. Each mis-tagged bunch shifts *all* of its hits by the tagging error.

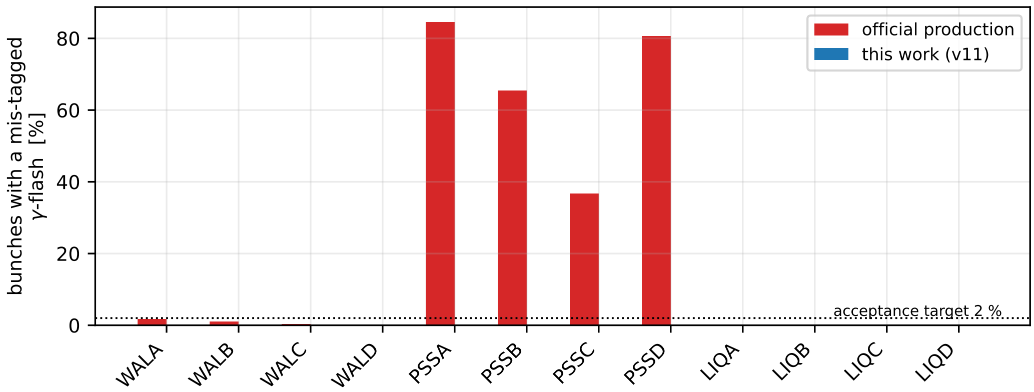

Figure 2: Fraction of bunches whose stored t_{flash} is more than 150 ns from that tree's own per-run mode, over 421 bunches. The liquids and the pickup are already clean, which is what shows the plastics' failure is configuration and not beam.

clamped copy of the real flash leaking through the closed gate at 11.60 μs; and the gate *opening* at 12.3 μs with a rail-to-rail transient whose baseline takes tens of microseconds to recover.

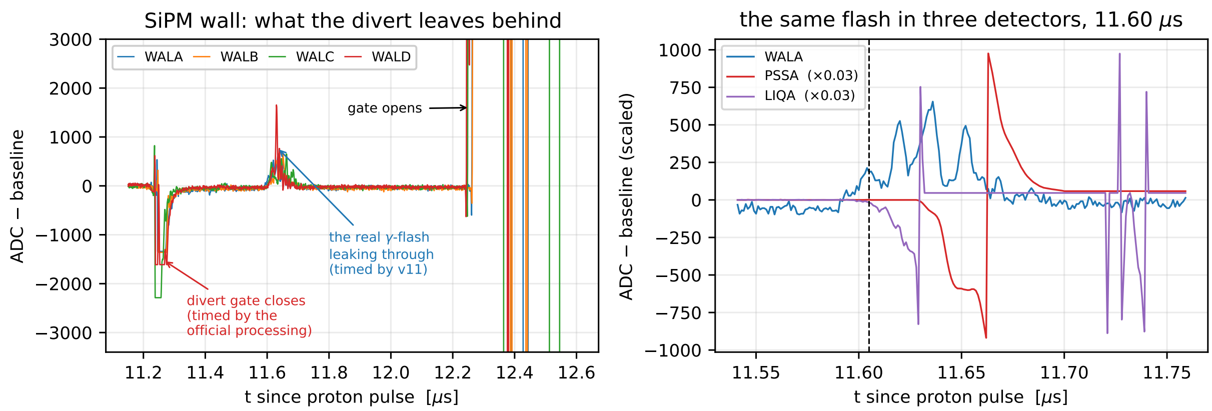

Figure 3: Raw stream1 waveforms, bunch 161. *Left*: one channel of each wall through the gate window. *Right*: the leak seen by the wall next to the same flash in a plastic and a liquid, neither of which is gated (scaled by 0.03). All three rise at 11.60 μs: the wall feature really is the flash.

With `G-FLASH THRESHOLD = 500.` and no time limit, which of the first two features wins is decided by their relative amplitude — and WALB's gate transient is the weakest of the four walls, so WALB usually lands on the flash while WALA/C/D land on the gate. **That bistability is the origin of the ~350 ns cross-detector offset**, and it is why arm B looked innocent while the other three did not.

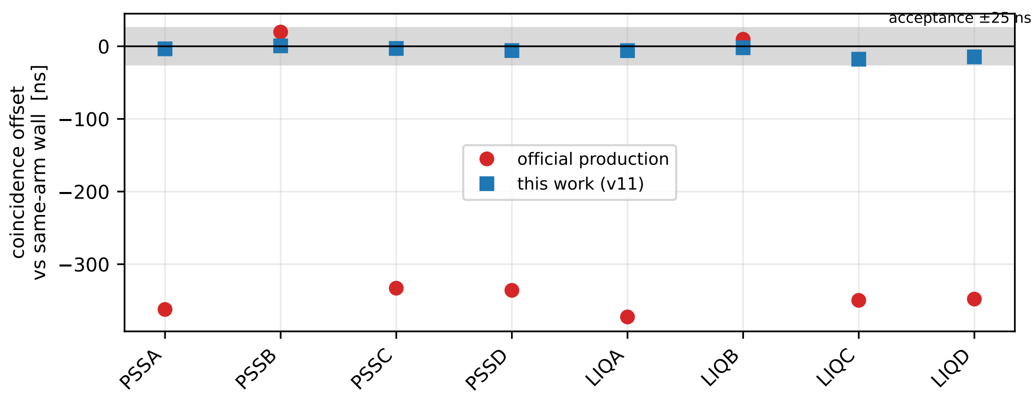

Figure 4: Position of the prompt coincidence peak between a large hit in each plastic or liquid and a hit in the same arm's wall. A genuine coincidence should sit at zero. In the official processing three arms out of four are displaced by the gate's lead time.

The fix is a lower time limit placed between the two features: `G-FLASH THRESHOLD = 250/11400.` The threshold must stay at or below 400: at 600 the weakest channels miss the leak entirely and fall through to the gate-opening transient, at which point the spread explodes from ~30 ns to ~660 ns.

What the divert costs regardless. After the gate reopens the wall baseline is +21 500 channels at 13 μs and still +400 at 30 μs. Wall hits are unusable from about 11.6 to 13–15 μs and degraded to ~40 μs— E_n above roughly 1 keV at the 19.5 m flight path. This is a property of the hardware, not of the configuration, but it belongs in any flux or E_n analysis.

4 Elimination windows that removed real pulses

Table 1: Area/amplitude of isolated late-time pulses, measured directly from the raw waveforms, against the shipped elimination window.

family	measured p1 … p99	shipped cut	fraction removed
WAL	42 … 110	10 … 200	0 %
PSS	1.3 … 29	2 … 20	∼25 %
LIQ	1.7 … 14	2 … 10	∼19 %

The wall cut sits a factor two above the largest value the walls ever produce, which is what a safe elimination window looks like; the plastic and liquid cuts sit inside the bulk of their own distributions (Fig. 5). Both were widened to 1 … 60. The PSA guide is explicit that elimination should be loose and that final rejection belongs in the offline analysis.

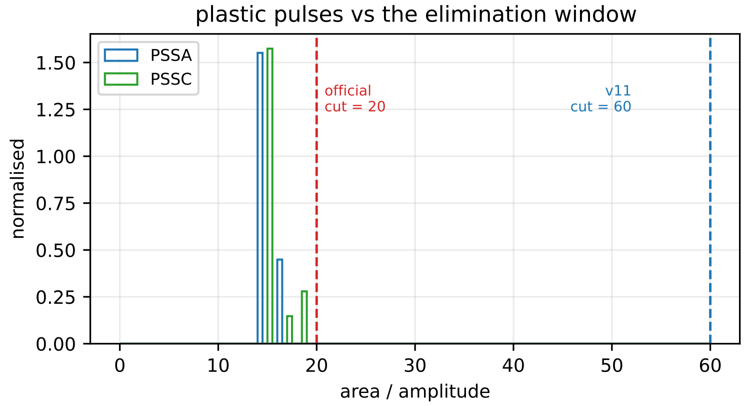

Figure 5: Area/amplitude of plastic pulses in the revised processing, with the old and new upper cuts marked. Everything to the right of the red line was being discarded.

5 Pulse shapes

5.1 The walls: the shipped templates are too short

Each shipped wall template is a *single raw pulse*, 314 ns long. The wall pulse is still at 4–6 % of its peak 200 ns after the maximum and 0.5 % at 500 ns, so the template ends inside the tail (Fig. 6). Every fitted amplitude then mis-accounts that tail, and the pileup deconvolution is working with a truncated kernel.

Table 2: Median fit χ^2 per wall, shipped vs measured templates.

	WALA	WALB	WALC	WALD
shipped (single pulse, 314 ns)	0.896	1.227	1.133	1.184
measured (median, 720–861 ns)	**0.851**	**1.002**	**1.002**	**1.062**

5.2 The plastics: fitting enabled, and it was the largest single gain

The plastics used `AMPLITUDE OPTION = 1` — a parabolic fit to the top, no deconvolution at all — while being the highest-rate tree in the file. Switching to shape fitting with measured 101 ns templates produced the biggest downstream improvement of any change, and by an instructive

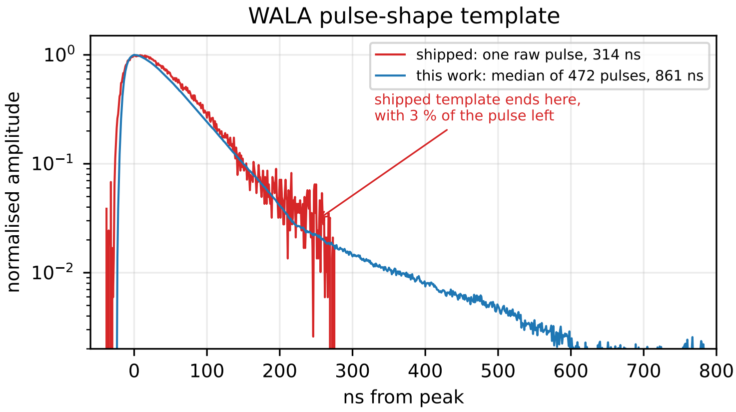

Figure 6: WALA pulse-shape template, normalised, logarithmic. The replacement is the median of 472 clean isolated pulses and runs to 861 ns.

route: it yields *fewer* plastic hits (0.72–0.99 of the previous count at every amplitude cut) but *better-timed* ones, merging pileup fragments back into a single correctly-timed pulse. The number of valid wall $\wedge$ plastic candidates rises even as the number of plastic hits falls.

5.3 The liquids: the templates were never the problem

We tried replacing the liquid templates three times — 551 ns averages, 81 ns averages, per-detector rather than shared — and every attempt was worse, with χ^2 more than doubling and amplitudes falling ∼30 %. A night on the raw waveforms explains it, and the explanation is not about templates at all.

They are a pileup problem. Only **8–24 % of liquid pulses are isolated** (LIQA 1014 of 6965 raw blocks, LIQB 812/10033, LIQC 136/1418, LIQD 1250/5175). Every template we built, and every test that said it was good, used that isolated minority. A longer, more faithful template overlaps more neighbours in a population that is mostly overlapping — which is exactly the 15–30 % hit loss we measured each time.

Single-pulse fit quality is floored by photon statistics. Fitting one measured template to isolated pulses, the residual scales as $\sqrt{A}$, not as A:

LIQD amplitude	1 900	3 700	5 100	7 500	14 000
residual / peak	0.0140	0.0103	0.0090	0.0077	0.0061
residual / $\sqrt{A}$	**0.61**	**0.62**	**0.64**	**0.65**	**0.67**

Flat to 10 % over a factor 25 in amplitude. The slow component is a countable number of photoelectrons and fluctuates irreducibly, so no template basis can absorb it — binning a basis by tail fraction moved χ^2 only from 70.8 to 63.1 over one to eight shapes. (For completeness: the hypothesis that a liquid carries two pulse classes, neutron-like and gamma-like, is *not* supported — above 3000 ADC the tail/total ratio is a single tight band at 0.21.)

What did work is `STEP SIZE` 2/4 → 1/3, the finest derivative window, for a 6 ns FWHM pulse at 1 GS/s: **+14 to +21 % yield** on all four liquids with χ^2 neutral-to-better (LIQD −8.5 %), and the pileup flag rate up ∼50 %.

What did not, and matters anyway. `afast` and `aslow` — the pulse-shape-discrimination observables, and the reason one runs a liquid scintillator — are **0.0 % filled** in the current processing. Setting the fast/slow boundary fills `afast` for 100 % of hits but leaves `aslow` at zero, because `aslow` is integrated "from the boundary up to the end of the pulse" and with `EXPAND PULSES = 0` the liquid pulse closes 20–40 ns after the peak while the slow component runs to ∼150 ns. Enabling expansion to reach it backfired: `aslow` stayed zero at the median and the expansion merged pulses in the pileup-dominated population, costing 17–28 % of the hits.

Consequence beyond PSD: the reported liquid `area` has been missing its slow component all along, in this and every previous processing. Anything calibrated on liquid `area` is affected. Recovering the slow component pulse-by-pulse appears to need the raw waveforms.

6 What the revision achieves, and what it costs

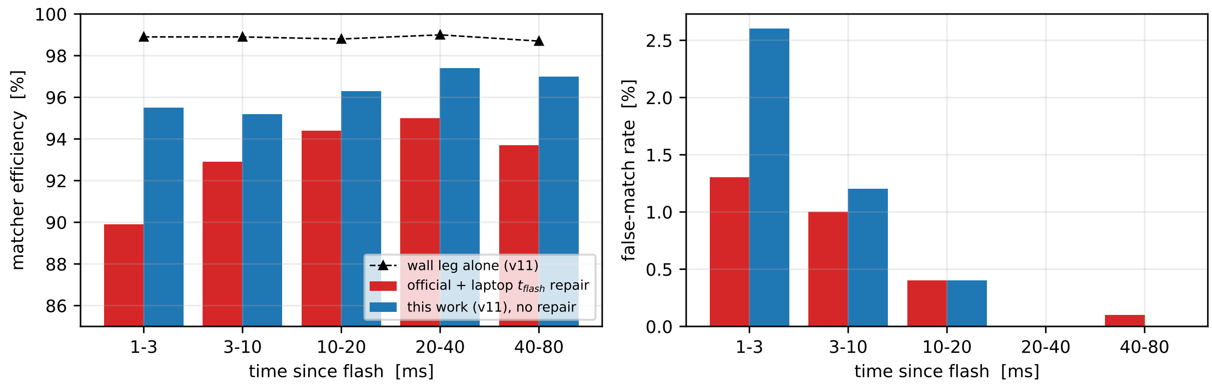

Figure 7: DREAM coincidence matcher, thresholded wall ∧ plastic singles. *Left*: efficiency per time-since-flash bin; the triangles are the wall leg alone, which shows the plastic leg is what limits the AND. *Right*: the false-match rate, measured with a +100 μs time-shifted control. The official column already includes our offline t_{flash} repair — with the raw official file the matcher is 12.7 % efficient and unusable.

Table 3: Summary on run 224572, DREAM run_79/stat090_0000. The revised column is measured with our offline t_{flash} repair *disabled*.

	official production	this revision
flash mis-identification	PSS 37–85 %	0.0 % on 12 of 13 trees
per-arm coincidence offset	−362/+20/−333/−336 ns	+1.5/+2.0/+1.0/−2.0 ns
matcher efficiency	93.7 % *(needs our repair)*	**96.4 %**
matcher false rate	0.5 %	0.6 %
in the 1–3 ms bin	89.9 % / 1.3 %	**95.5 %** / 2.6 %
wall timing, top↔bottom σ	6.65 ns → 6.65 ns	
wall↔plastic coincidence σ	6.46 ns → 6.41 ns	
MIP peak width (FWHM/peak)	1.22 → 1.22	
timing walk over amplitude	1.38 ns → 1.48 ns	

The lower half of that table is the guard: it would be easy to gain hits by reconstructing them worse, so timing resolution, amplitude linearity and MIP width were tracked alongside efficiency. All are unchanged within a few percent.

The cost, stated plainly. The false-match rate in the 1–3 ms bin roughly doubles, from 1.3 % to 2.6 %. That is the direct consequence of recovering the plastic hits: the candidate rate rises from ∼935 to ∼1042 per bunch. For this analysis it is a good trade, because efficiency in the same bin rises by 5.6 points and accidentals can be subtracted offline with a time-shifted control. An analysis that needs early-time purity instead can trade it back by tightening `AREA/AMP HIGH` towards 20.

Where the limit now is. The wall leg alone reaches 98.9 % and is flat in time; the AND reaches 96.4 %. The plastic leg therefore still costs 2.5 % overall and 3.4–3.7 % at early times. We established that this is *not* a pulse-recognition problem: three further variants with materially different plastic reconstructions — ±40 % in amplitude, ±30 % in hit count — all return exactly 96.4 %, because above ∼2000 ADC they agree to better than 1 % and the hardware discriminator never sees anything below that. Further gains are analysis-side: the per-arm discriminator threshold model, the channel selection, dead time, or real detector inefficiency.

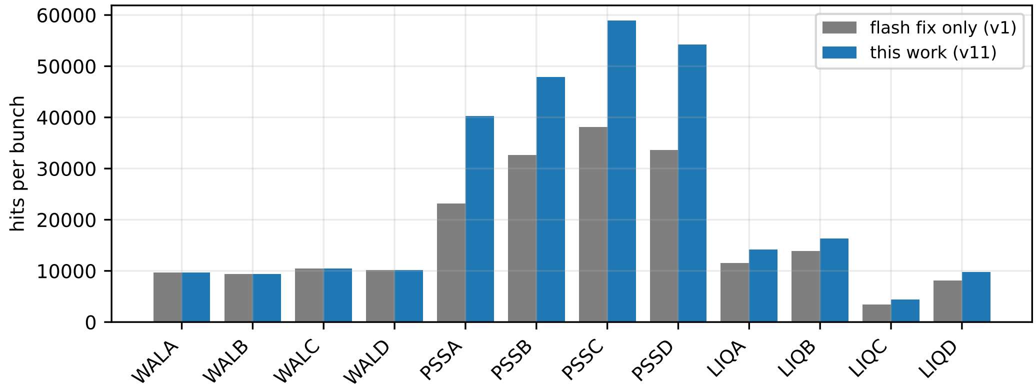

Figure 8: Hits per bunch per tree, flash fix alone versus the full revision. The walls are unchanged by construction — their elimination windows were already safe — which is the control that the change did what it was meant to and nothing else.

7 Reproducing this

All tooling is in the group repository under `ntof_processing/`: `grade_candidate.py` (flash identification, coincidence offsets, hit counts), `compare_fits.py` (fit χ^2), `quality_metrics.py` (timing resolution, amplitude linearity, MIP width, timing walk — all accidental-subtracted), `dream_regression.py` (the matcher), `make_variants.py` (generates a UserInput as a base plus named column edits), `flash_finder_emulator.py` (pre-validates G-FLASH parameters on raw waveforms without a processing round-trip), and `make_pulse_shapes.py` (builds the templates). The numbers quoted here are in `report/results.json` with the tool that produced each one.

Two traps worth passing on. Coincidence widths must be accidental-subtracted: at these rates the nearest-hit-in-window distribution is dominated by chance, and without an off-time sideband subtracted the wall↔plastic width reads 38.8 ns instead of 6.4 ns. And the merge step of `RunProcessing.sh` failed on every EAR2 run we tried, with `max total download bytes exceeded (max=1024 MB)` — the per-file processing always succeeded, so we read the partials in `completed/` directly.